

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<title>Advanced GPA Calculator</title>

<meta name="viewport" content="width=device-width, initial-scale=1.0">


<!-- Chart.js -->

<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

<!-- jsPDF -->

<script src="https://cdnjs.cloudflare.com/ajax/libs/jspdf/2.5.1/jspdf.umd.min.js"></script>


<style>

body {

    background: #121212;

    color: #ffffff;

    font-family: Arial, sans-serif;

    padding: 20px;

}

.box {

    background: #1e1e1e;

    padding: 20px;

    max-width: 480px;

    margin: auto;

    border-radius: 10px;
```

```
}
```

```
h2 {
```

```
  text-align: center;
```

```
}
```

```
.course {
```

```
  display: grid;
```

```
  grid-template-columns: 1.3fr 1fr 0.7fr 0.3fr;
```

```
  gap: 6px;
```

```
  margin-bottom: 8px;
```

```
}
```

```
input, select {
```

```
  padding: 8px;
```

```
  background: #2c2c2c;
```

```
  color: white;
```

```
  border: 1px solid #444;
```

```
  border-radius: 5px;
```

```
}
```

```
button {
```

```
  width: 100%;
```

```
  padding: 10px;
```

```
  margin-top: 8px;
```

```
  background: #0d6efd;
```

```
color: white;
border: none;
border-radius: 6px;
font-size: 15px;
cursor: pointer;
}
```

```
button:hover {
  opacity: 0.9;
}
```

```
#result {
  margin-top: 12px;
  text-align: center;
  font-weight: bold;
}
```

```
.warning {
  margin-top: 10px;
  text-align: center;
  color: #ff5555;
  font-size: 14px;
}
```

```
canvas {
  margin-top: 15px;
```

```
}  
  
</style>  
  
</head>  
  
<body>  
  
<div class="box">  
  
<h2>🏠 GPA Calculator</h2>  
  
<select id="scale">  
  <option value="4">4.0 Scale</option>  
  <option value="5">5.0 Scale</option>  
</select>  
  
<div id="courses">  
  <div class="course">  
    <input placeholder="Subject Name">  
    <input placeholder="Grade (A, B+, B-)">  
    <input type="number" placeholder="Cr">  
    <input type="checkbox" checked="" title="Include">  
  </div>  
</div>  
  
<button onclick="addCourse()">Add Course</button>  
<button onclick="calculate()">Calculate GPA</button>  
<button onclick="saveData()">Save Data</button>
```

```
<button onclick="exportPDF()">Export PDF</button>
```

```
<div id="result"></div>
```

```
<div class="warning"> ⚠ /> أوعي تشيل مادة</div>
```

```
<canvas id="chart"></canvas>
```

```
</div>
```

```
<script>
```

```
const courses = document.getElementById("courses");
```

```
const result = document.getElementById("result");
```

```
const scaleSelect = document.getElementById("scale");
```

```
let chart;
```

```
// GPA scales
```

```
function getGrades(scale) {
```

```
  if (scale == 4) {
```

```
    return {
```

```
      "A":4,"A-":3.7,
```

```
      "B+":3.3,"B":3,"B-":2.7,
```

```
      "C+":2.3,"C":2,"C-":1.7,
```

```
      "D+":1.3,"D":1,
```

```
      "F":0
```

```
    };
```

```
  }
```

```
  return {
```

```
"A":5,"A-":4.5,  
"B+":4,"B":3.5,"B-":3,  
"C+":2.5,"C":2,  
"D":1,"F":0  
};  
}
```

```
// Add course
```

```
function addCourse() {  
  courses.innerHTML += `  
    <div class="course">  
      <input placeholder="Subject Name">  
      <input placeholder="Grade (A, B+, B-)">  
      <input type="number" placeholder="Cr">  
      <input type="checkbox" checked>  
    </div>`;   
}
```

```
// Calculate GPA
```

```
function calculate() {  
  const grades = getGrades(scaleSelect.value);  
  let totalPoints = 0;  
  let totalCredits = 0;  
  let distribution = {};  
  let errors = [];
```

```
document.querySelectorAll(".course").forEach((c, i) => {  
  const subject = c.children[0].value.trim();  
  const grade = c.children[1].value.toUpperCase().trim();  
  const credits = Number(c.children[2].value);  
  const included = c.children[3].checked;  
  
  if (!included) return;  
  
  if (!subject) {  
    errors.push(` Course ${i+1}: Subject name missing`);  
    return;  
  }  
  
  if (!(grade in grades)) {  
    errors.push(` Course ${i+1}: Invalid grade`);  
    return;  
  }  
  
  if (credits <= 0 || isNaN(credits)) {  
    errors.push(` Course ${i+1}: Invalid credits`);  
    return;  
  }  
  
  totalPoints += grades[grade] * credits;  
  totalCredits += credits;  
  distribution[grade] = (distribution[grade] || 0) + 1;  
});
```

```
if (errors.length > 0) {  
  result.innerHTML = `  
    <div style="color:#ff5555">  
      ❌ Fix the following:  
      <ul>${errors.map(e => `<li>${e}</li>`).join("")}</ul>  
    </div>`;   
  return;  
}
```

```
if (totalCredits === 0) {  
  result.innerHTML = "⚠️ No courses selected";  
  return;  
}
```

```
const gpa = totalPoints / totalCredits;
```

```
const message =
```

```
gpa >= 3.8 ? "🏆 يا دحيح! عاش يا بطل"
```

```
gpa >= 3.0 ? "👏 مستوى عالي كمل"
```

```
gpa >= 2.0 ? "😊 شد حيلك شوية"
```

```
"الحق نفسك قبل" Probation 🚫";
```

```
result.innerHTML = `
```

```
GPA: ${gpa.toFixed(2)}<br>
```

```
CGPA: ${gpa.toFixed(2)}<br>
```



```
    ${message}
```

```
`;
```

```
drawChart(distribution);
```

```
}
```

```
// Draw chart
```

```
function drawChart(data) {
```

```
    if (chart) chart.destroy();
```

```
    chart = new Chart(document.getElementById("chart"), {
```

```
        type: "pie",
```

```
        data: {
```

```
            labels: Object.keys(data),
```

```
            datasets: [{
```

```
                data: Object.values(data)
```

```
            ]
```

```
        }
```

```
    });
```

```
}
```

```
// Save data
```

```
function saveData() {
```

```
    const data = [];
```

```
    document.querySelectorAll(".course").forEach(c => {
```

```
        data.push({
```

```
            s: c.children[0].value,
```

```
    g: c.children[1].value,  
    cr: c.children[2].value,  
    i: c.children[3].checked  
  });  
});  
localStorage.setItem("gpaData", JSON.stringify(data));  
alert("✅ Data Saved");  
}
```

```
// Load data
```

```
window.onload = () => {  
  const data = JSON.parse(localStorage.getItem("gpaData") || "[]");  
  if (!data.length) return;  
  courses.innerHTML = "";  
  data.forEach(d => {  
    courses.innerHTML += `  
    <div class="course">  
      <input value="${d.s}">  
      <input value="${d.g}">  
      <input type="number" value="${d.cr}">  
      <input type="checkbox" ${d.i ? "checked" : ""}>  
    </div>`;  
  });  
};
```

```
// Export PDF
```

```
function exportPDF() {  
  const { jsPDF } = window.jspdf;  
  const pdf = new jsPDF();  
  pdf.text("GPA Report", 20, 20);  
  pdf.text(result.innerText || "No data", 20, 40);  
  pdf.save("GPA_Report.pdf");  
}  
</script>  
  
</body>  
</html>
```